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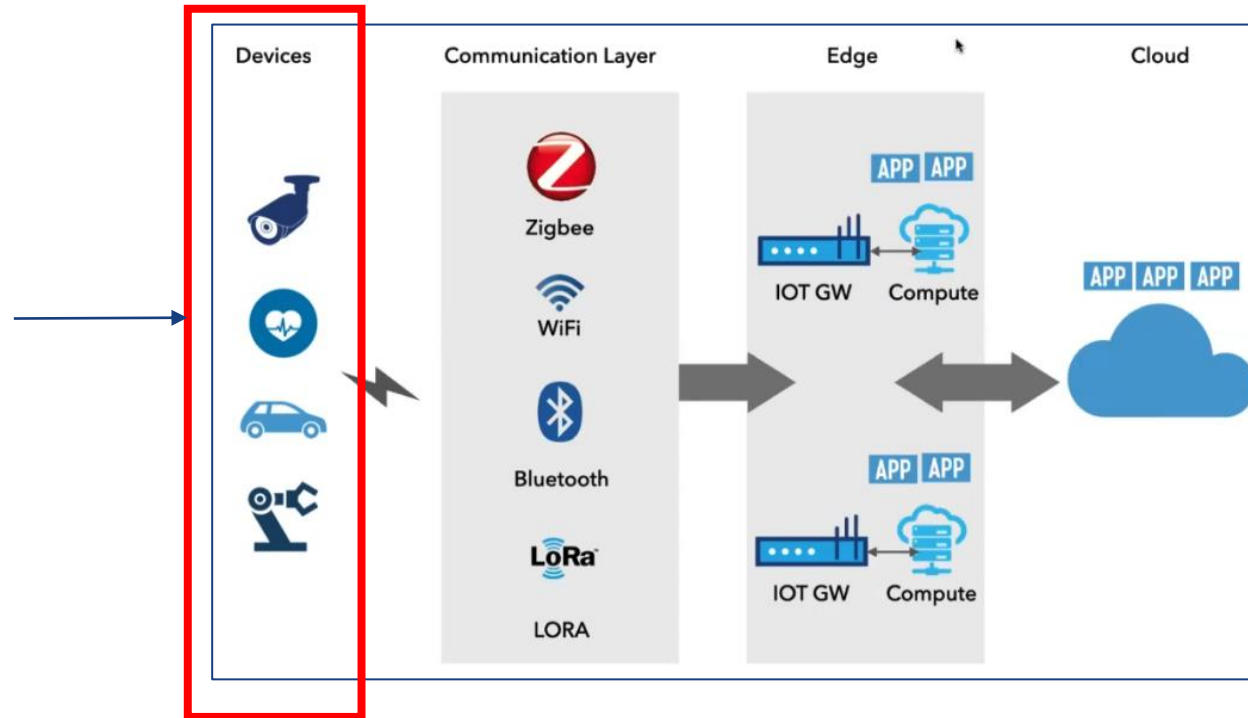


WORKSHOP DAY 1

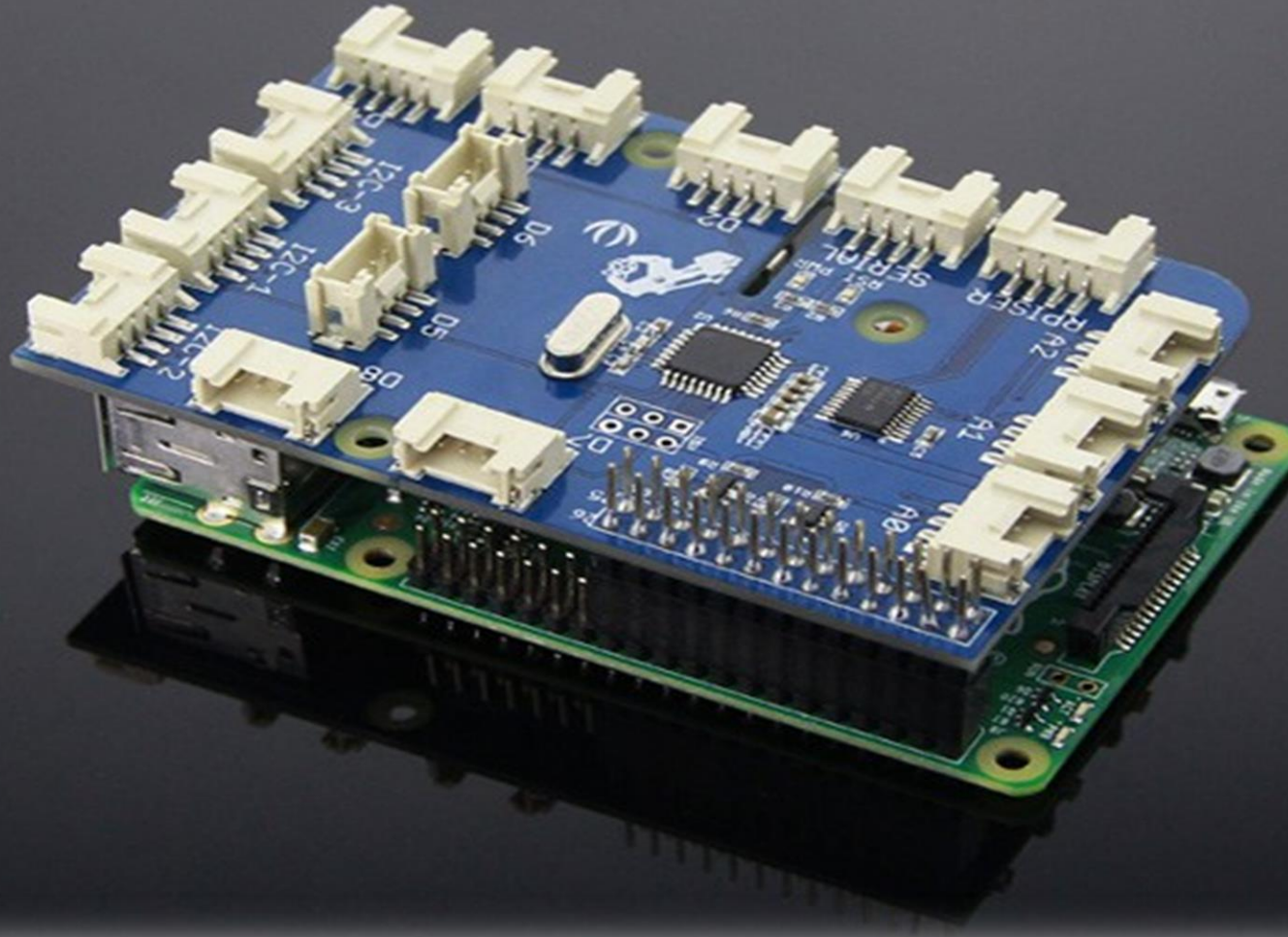
Workshop Themes

- Day 1:
 - Get familiar with RPi and GrovePi+
 - Learn how to interface with sensors and actuators using Python
 - Learn how to build a simple smart system, focusing on the edge device layer:

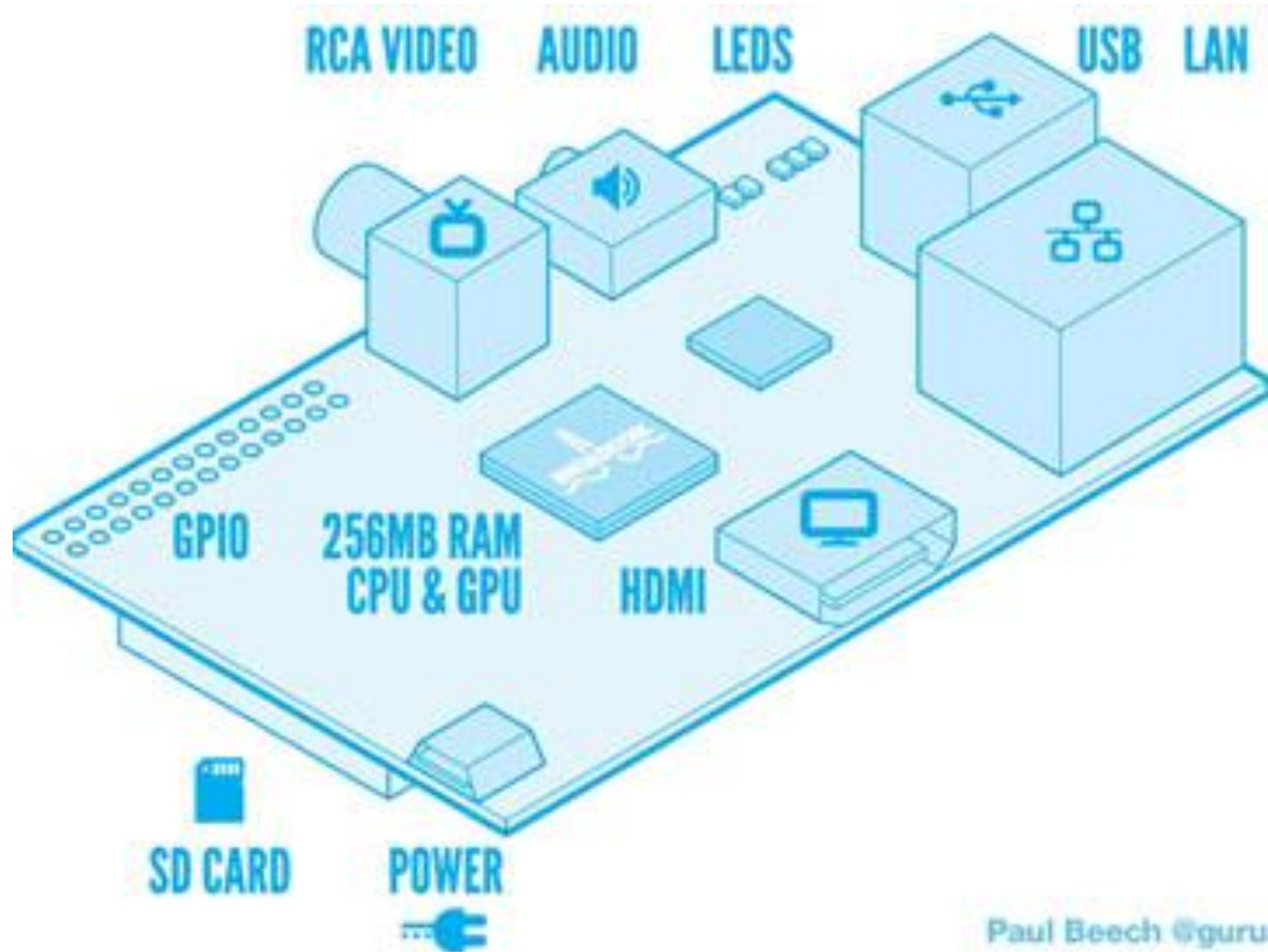
We will be focusing on this part on Day 1 Workshop



Raspberry Pi with GrovePi+, Sensors & Camera



Introduction to Raspberry PI

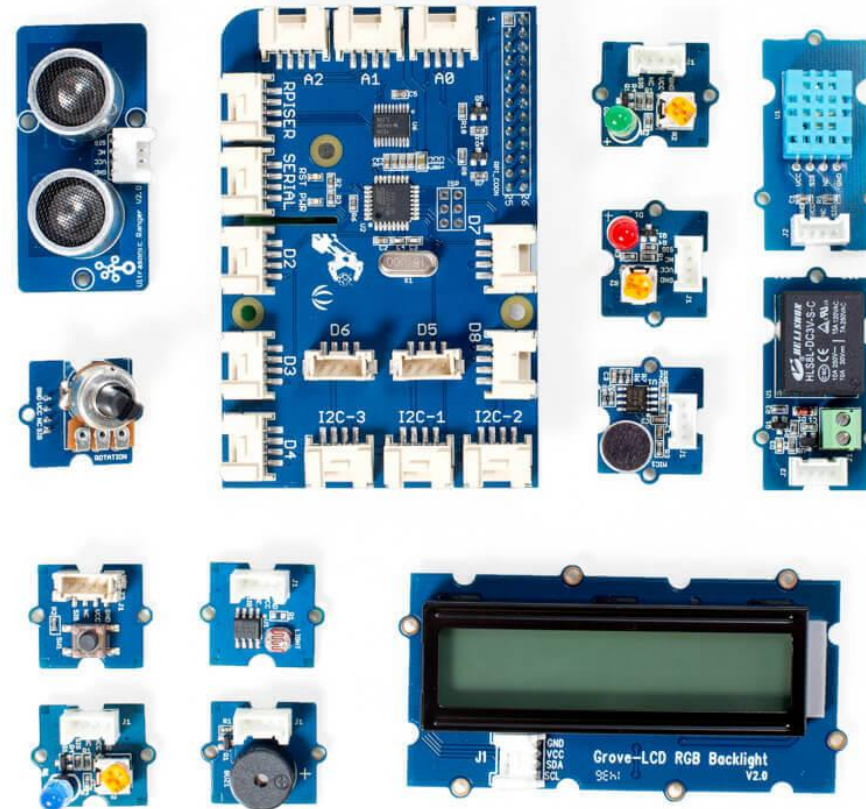


Various components:

1. **ARM CPU/GPU**
2. **GPIO** (General-purpose input/output connection)
3. **RCA** (allows connection of analog TVs)
4. **Audio out**
5. **LEDs**
6. **USB**
7. **HDMI**
8. **Power**
9. **SD cardslot**
10. **Ethernet**

Introduction to Raspberry PI

- An IoT device
- **Low cost, credit-card sized computer**
- Capable of doing everything a desktop computer can do
- Ability to interact with the outside world
- Built-in wireless/bluetooth connectivity



Introduction to GrovePi+ Starter Kit



Introduction to GrovePi+ Starter Kit



Many sensors selection:

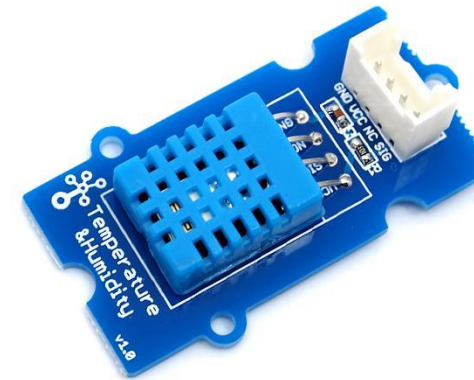
1. **Temperature**
2. **Humidity**
3. **Light**
4. **Sound**
5. **Ultrasonic**
6. **Relay**

Output selection:

1. **LED**
2. **Buzzer**

Understanding Sensors

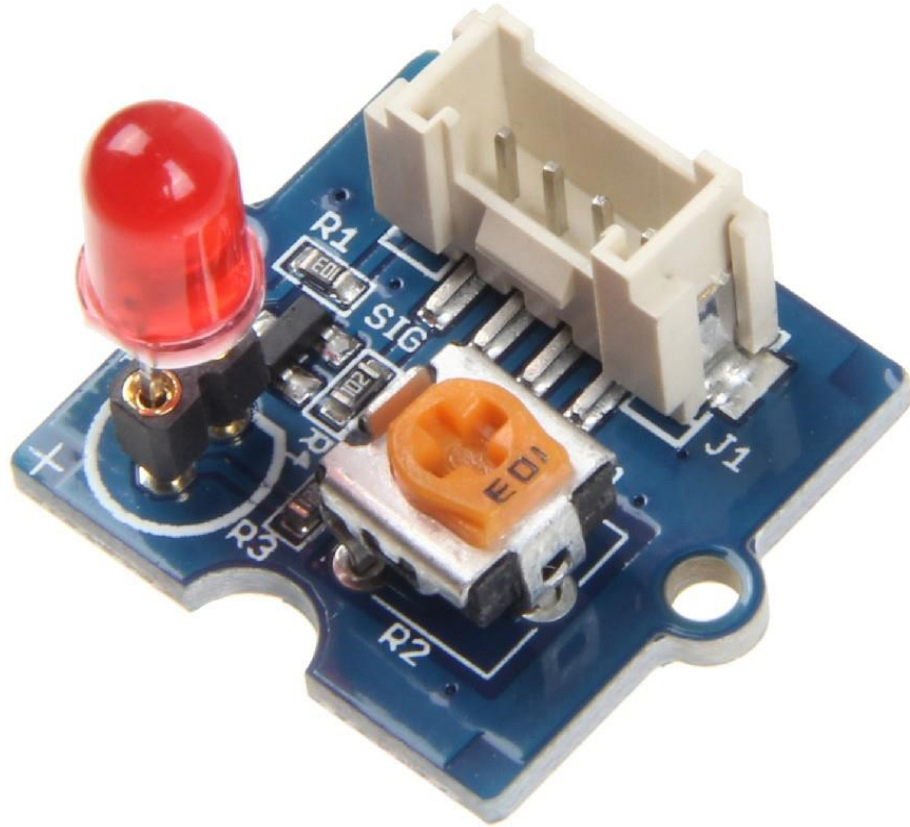
- Light sensor – Analog
- Sound sensor – Analog
- Temperature & Humidity (DHT) Sensor – Digital



Getting familiar with GrovePi+, Sensors & Camera

1. LED light indication (Output)
2. Buzzer (Output)
3. Button Sensor
4. DHT Temperature & Humidity Sensor
5. Light Sensor
6. Sound Sensor
7. Ultrasonic Sensor
8. Camera

1. LED light indication (Output)



1. Connect Grove-Red Led to **D3 port** of GrovePi+.
2. Open the **grove_led_blink.py** python script located on /Dexter/GrovePi/Software/Python
3. Change LED = 4 to LED = 3 as the LED is connected to digital port 3. Try to understand the code and how it works

1. LED light indication (Output)

```
import time

from grovepi import *

# Connect the Grove LED to digital port D3 instead of D4
led = 3

pinMode(led, "OUTPUT")
time.sleep(1)

print ("This example will blink a Grove LED connected to the
GrovePi+ on the port D().\nIf you're having trouble seeing the LED
blink, be sure to check the LED connection and the port number.\nYou
may also try reversing the direction of the LED on the
sensor.".format(led))

while True:
    try:

        #Blink the LED
        digitalWrite(led,1)      # Send HIGH to switch on LED
        print ("LED ON!")

        digitalWrite(led,0)      # Send LOW to switch off LED
        print ("LED OFF!")

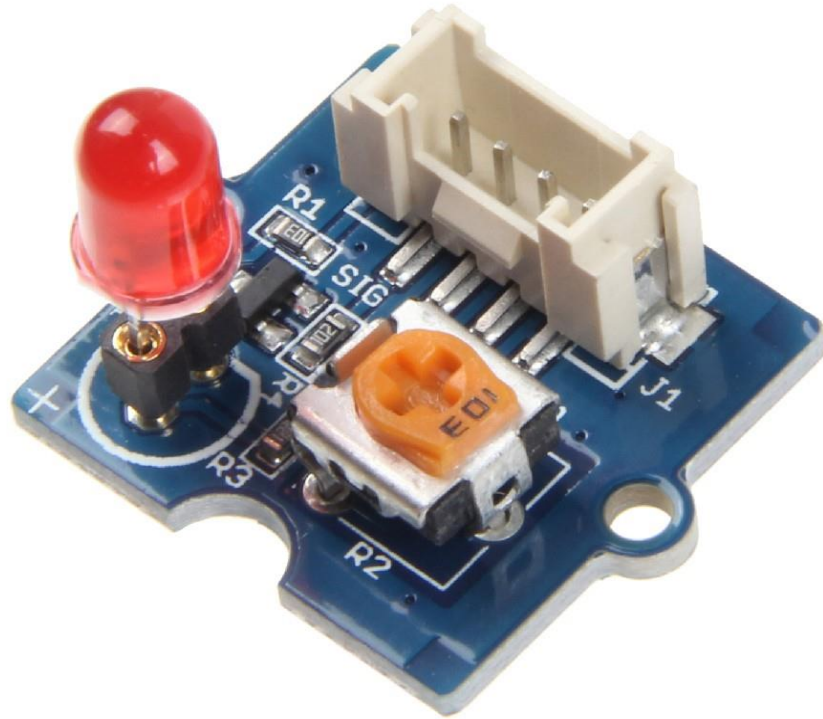
        time.sleep(0.5)

    except KeyboardInterrupt: # Turn LED off before stopping
        digitalWrite(led,0)

        break

    except IOError:          # Print "Error" if communication error
        encountered
```

1. LED light indication (Output)

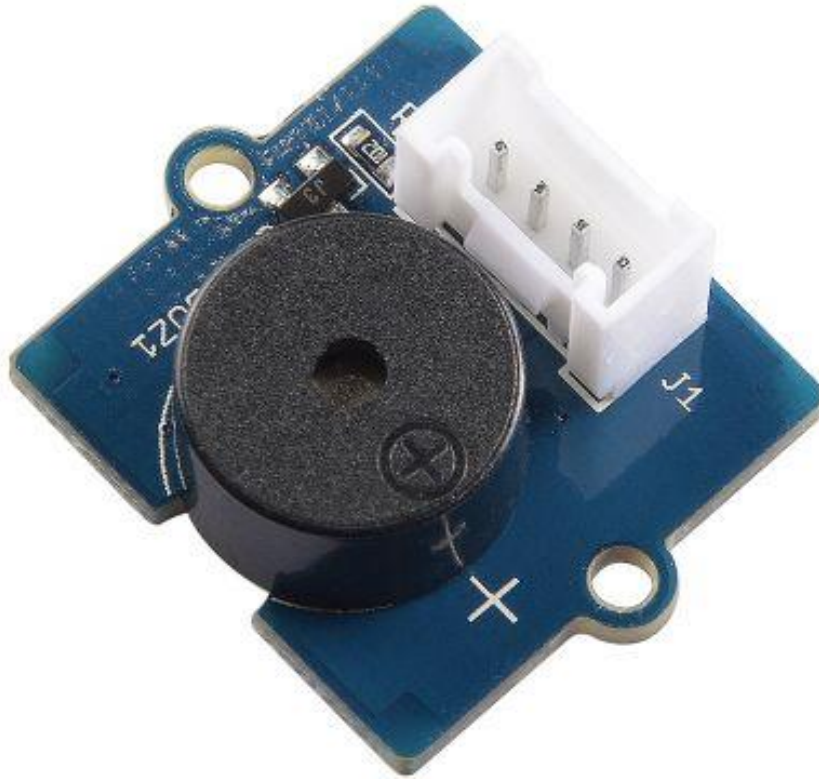


4. Execute following commands and you will see the LED on and off:

```
$ cd ~/Dexter/GrovePi/Software/Python
```

```
$ python3 grove_led_blink.py
```

2. Buzzer (Output)



1. Connect Grove-Buzzer to **D8 port** of GrovePi+
2. Open the **grove_buzzer.py** python script located on /Dexter/GrovePi/Software/Python
3. Try to understand the code and how it works

2. Buzzer (Output)

```
import time
import grovepi

# Connect the Grove Buzzer to digital port D8
# SIG, NC, VCC, GND

buzzer = 8

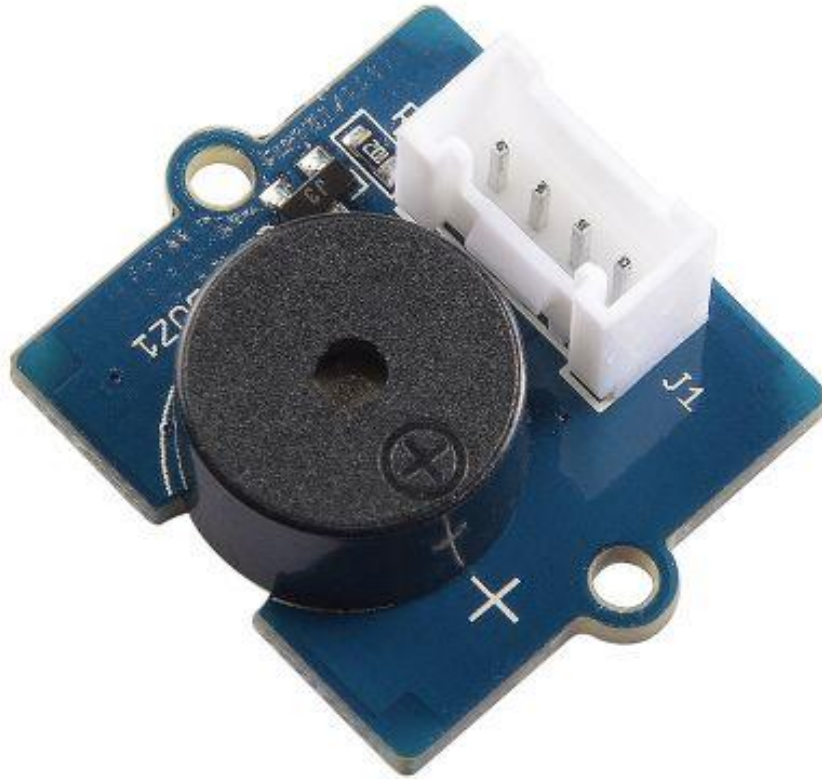
grovepi.pinMode(buzzer, "OUTPUT")
```

```
while True:
    try:
        # Buzz for 1 second
        grovepi.digitalWrite(buzzer, 1)
        print ('start')
        time.sleep(1)

        # Stop buzzing for 1 second and repeat
        grovepi.digitalWrite(buzzer, 0)
        print ('stop')
        time.sleep(1)

    except KeyboardInterrupt:
        grovepi.digitalWrite(buzzer, 0)
        break
    except IOError:
        print ("Error")
```

2. Buzzer (Output)



4. Execute following commands and you will hear the buzzer on and off:

```
$ cd ~/Dexter/GrovePi/Software/Python
```

```
$ python3 grove_buzzer.py
```

3. Button Sensor

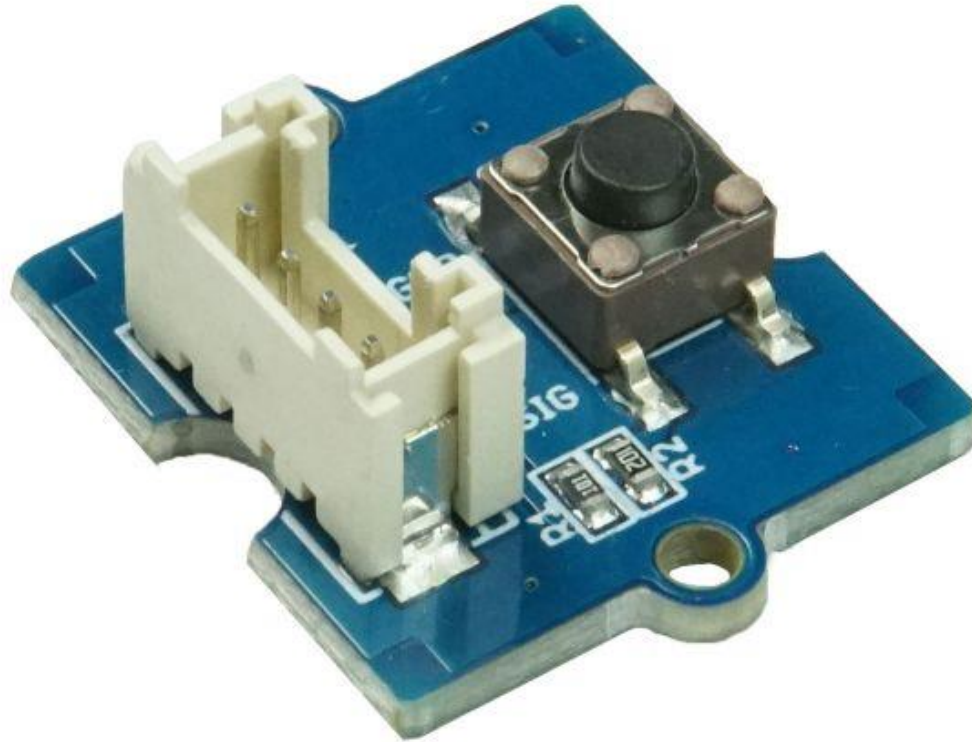


1. Connect Grove Button to **D3 port** of GrovePi+
2. Open the **grove_button.py** python script located on
/GrovePi/Software/Python
3. Try to understand the code and how it works

3. Button Sensor

```
1  import time
2  import grovepi
3
4  # Connect the Grove Button to digital port D3
5  # SIG,NC,VCC,GND
6  button = 3
7
8  grovepi.pinMode(button,"INPUT")
9
10 while True:
11     try:
12         print(grovepi.digitalRead(button))
13         time.sleep(.5)
14
15     except IOError:
16         print ("Error")
```

3. Button Sensor

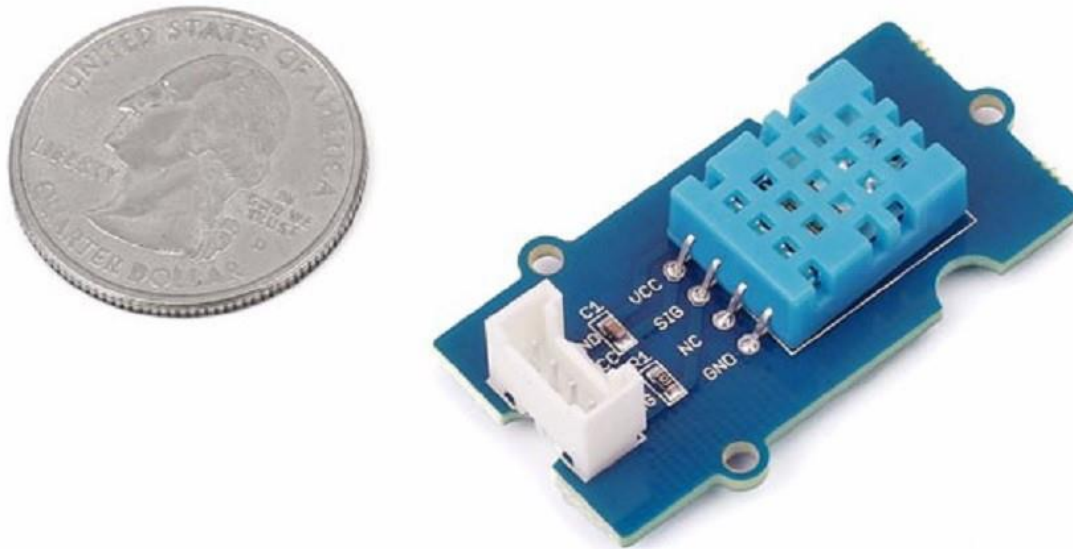


4. Execute following command;
press/release the button to see what
happens:

```
$ cd ~/Dexter/GrovePi/Software/Python
```

```
$ python3 grove_button.py
```

4. DHT Temperature & Humidity Sensor



1. Connect the Temperature & Humidity sensor to **D4 port** of GrovePi+
2. Open the **grove_dht_pro.py** python script located on /Dexter/GrovePi/Software/Python
3. Try to understand the code and how it works

4. DHT Temperature & Humidity Sensor

```
import grovepi
import math

# Connect the Grove Temperature & Humidity Sensor Pro to digital
port D4

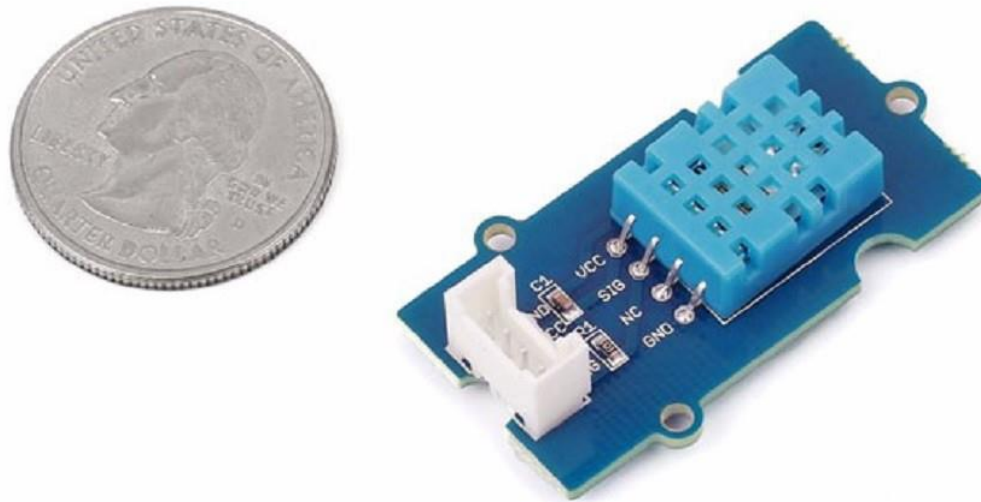
# This example uses the blue colored sensor.
# SIG,NC,VCC,GND
sensor = 4 # The Sensor goes on digital port 4.

# temp_humidity_sensor_type
# Grove Base Kit comes with the blue sensor.
blue = 0 # The Blue colored sensor.
white = 1 # The White colored sensor.

while True:
    try:
        # This example uses the blue colored sensor.
        # The first parameter is the port, the second parameter is
        the type of sensor.
        [temp,humidity] = grovepi.dht(sensor,blue)
        if math.isnan(temp) == False and math.isnan(humidity) ==
        False:
            print("temp = %.02f C humidity =%.02f%%"%(temp,
            humidity))

        except IOError:
            print ("Error")
```

4. DHT Temperature & Humidity Sensor



4. Execute following commands and you will see the temperature and humidity value printed on the terminal:

```
$ cd ~/Dexter/GrovePi/Software/Python
```

```
$ python3 grove_dht_pro.py
```

5. Light Sensor



1. Connect Grove **Light sensor** to **A0 port** of GrovePi+ and connect Grove **LED** to **D4**
2. Open the **grove_light_sensor.py** python script located on
/Dexter/GrovePi/Software/Python
3. Try to understand the code and how it works

5. Light Sensor

```
1 import time
2 import grovepi
3
4 # Connect the Grove Light Sensor to analog port A0
5 # SIG,NC,VCC,GND
6 light_sensor = 0
7
8 # Connect the LED to digital port D4
9 # SIG,NC,VCC,GND
10 led = 4
11
12 # Turn on LED once sensor exceeds threshold resistance
13 threshold = 10
14
15 grovepi.pinMode(light_sensor,"INPUT")
16 grovepi.pinMode(led,"OUTPUT")
17
```

```
18 while True:
19     try:
20         # Get sensor value
21         sensor_value = grovepi.analogRead(light_sensor)
22
23         # Calculate resistance of sensor in K
24         resistance = (float)(1023 - sensor_value) * 10 / sensor_value
25
26         if resistance > threshold:
27             # Send HIGH to switch on LED
28             grovepi.digitalWrite(led,1)
29         else:
30             # Send LOW to switch off LED
31             grovepi.digitalWrite(led,0)
32
33         print("sensor_value = %d resistance = %.2f" %(sensor_value, resistance))
34         time.sleep(.5)
35
36     except IOError:
37         print ("Error")
```

5. Light Sensor

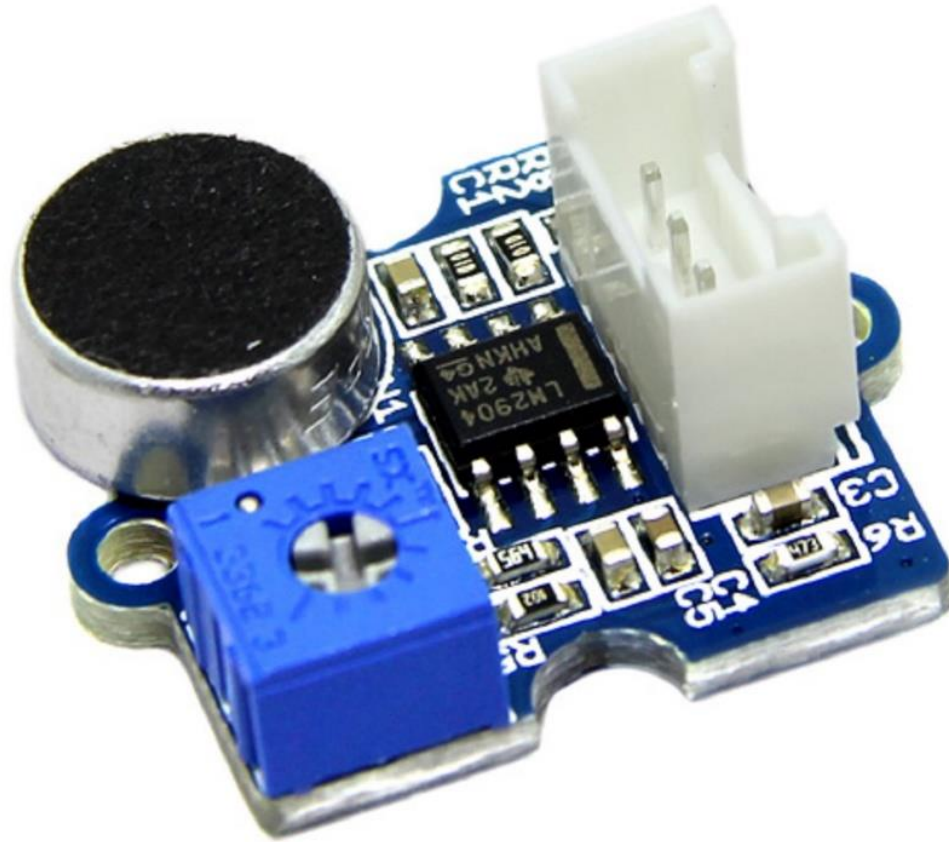


4. Execute following commands to obtain brightness values; cover (or shine bright light on) the light sensor and see what happens:

```
$ cd ~/Dexter/GrovePi/Software/Python
```

```
$ python3 grove_light_sensor.py
```

6. Sound Sensor

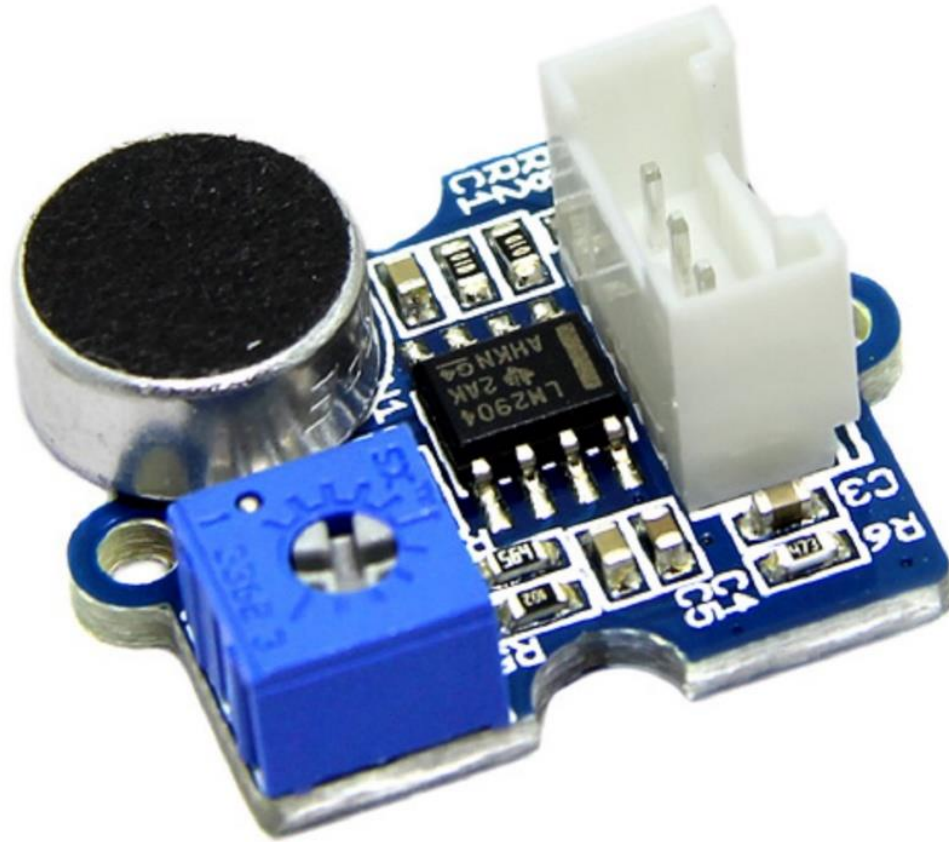


1. Connect Grove **Sound to A0** port of GrovePi+ and connect Grove **LED to D5**
2. Open the **grove_sound_sensor.py** python script located on /Dexter/GrovePi/Software/Python
3. Try to understand the code and how it works

6. Sound Sensor

```
35 import time
36 import grovepi
37
38 # Connect the Grove Sound Sensor to analog port A0
39 # SIG,NC,VCC,GND
40 sound_sensor = 0
41
42 # Connect the Grove LED to digital port D5
43 # SIG,NC,VCC,GND
44 led = 5
45
46 grovepi.pinMode(sound_sensor,"INPUT")
47 grovepi.pinMode(led,"OUTPUT")
48
49 # The threshold to turn the led on  $400.00 * 5 / 1024 = 1.95$ 
50 threshold_value = 400
51
52 while True:
53     try:
54         # Read the sound level
55         sensor_value = grovepi.analogRead(sound_sensor)
56
57         # If loud, illuminate LED, otherwise dim
58         if sensor_value > threshold_value:
59             grovepi.digitalWrite(led,1)
60         else:
61             grovepi.digitalWrite(led,0)
62
63         print("sensor_value = %d" %sensor_value)
64         time.sleep(.5)
65
66     except IOError:
67         print ("Error")
```

6. Sound Sensor



4. Execute following commands to obtain sound values; use your nail, and lightly and repeatedly knock it against the black area of the sound sensor; see what happens:

```
$ cd ~/Dexter/GrovePi/Software/Python
```

```
$ python3 grove_sound_sensor.py
```

7. Ultrasonic Sensor



1. Connect Grove **Ultrasonic Sensor to D4 port** of GrovePi+
2. Open the **grove_ultrasonic.py** python script located on /Dexter/GrovePi/Software/Python
3. Try to understand the code and how it works

7. Ultrasonic Sensor

```
1  # GrovePi + Grove Ultrasonic Ranger
2
3  from grovepi import *
4
5  # Connect the Grove Ultrasonic Ranger to digital port D4
6  # SIG,NC,VCC,GND
7
8  ultrasonic_ranger = 4
9
10 while True:
11     try:
12         # Read distance value from Ultrasonic
13         print ultrasonicRead(ultrasonic_ranger)
14
15     except TypeError:
16         print "Error"
17     except IOError:
18         print "Error"
```

7. Ultrasonic Sensor



4. Execute following commands to obtain distance values; put a solid object (e.g. your hand) in front of the ultrasonic sensor and see what happens:

```
$ cd ~/Dexter/GrovePi/Software/Python
```

```
$ python3 grove_ultrasonic.py
```

8. Camera



1. Connect the camera to **Raspberry Pi** (you may need to remove the grove shield first)
2. Open the **camera.py** python script located on ~/Desktop (if you do not have this file, please download it from Luminus)
3. Try to understand the code and how it works

8. Camera

```
from picamera import PiCamera
from time import sleep

camera = PiCamera()

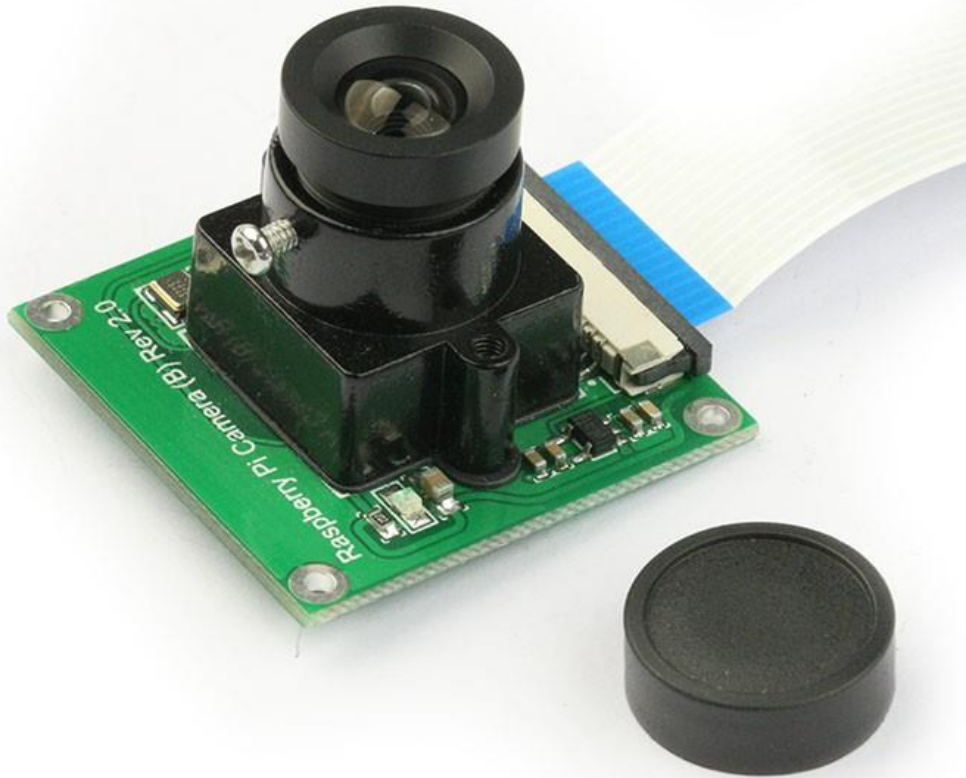
# The camera preview only works when a monitor is connected to your
Raspberry Pi. If you are using remote access (such as SSH or VNC),
you won't able to see the camera preview.
camera.start_preview()

# If your preview is upside-down, you can rotate it by 180 degrees
with the following code:
camera.rotation = 180

sleep(5)
camera.stop_preview()

# To save the captured image
camera.capture('/home/pi/Desktop/image.jpg')
```

8. Camera

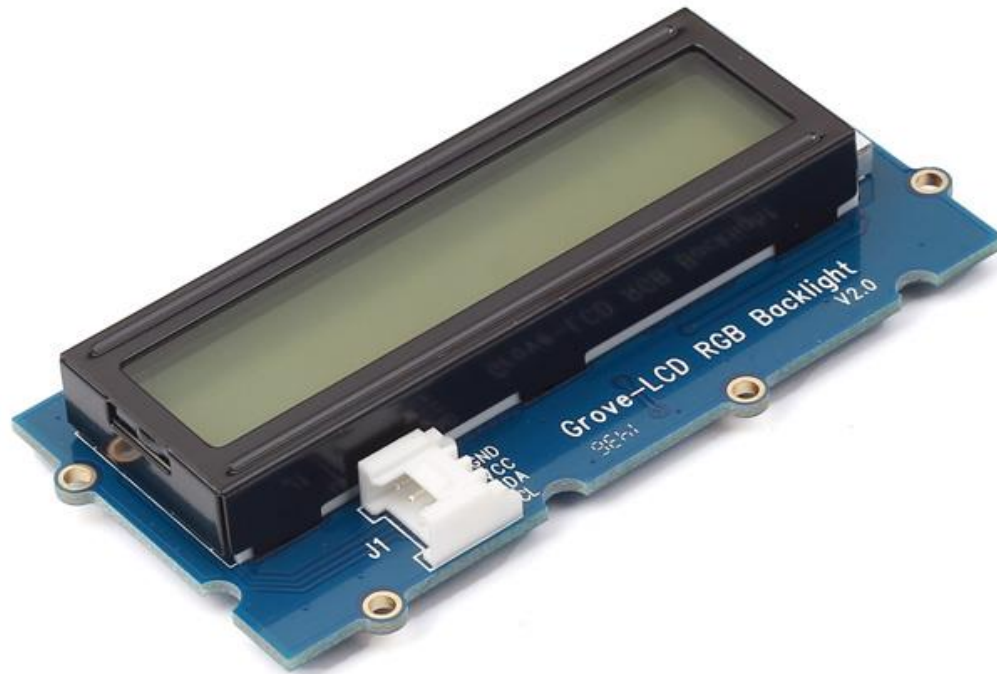


4. Execute following commands and you will see the captured image on your desktop:

```
$ cd ~/Desktop
```

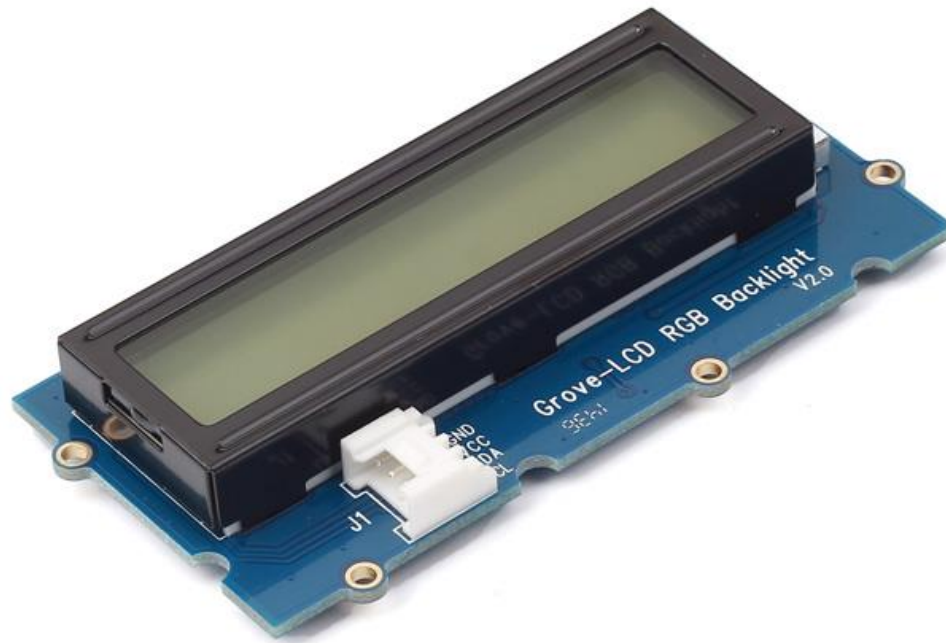
```
$ sudo python3 camera.py
```

9. LCD RGB Backlight (Output)



1. Connect Grove **LCD RGB** to **I2C-1 port** of GrovePi+
2. Open the **example.py** python script located on
/Dexter/GrovePi/Software/Python/
grove_rgb_lcd
3. Try to understand the code and how it works

9. LCD RGB Backlight (Output)



4. Execute following commands and you will observe some wordings will appear in the screen:

```
$ cd ~/Dexter/GrovePi/Software/Python  
/grove_rgb_lcd
```

```
$ python3 example.py
```

OR

```
$ python3 example2.py
```

OR

```
$ python3 example3.py
```

Project 1: Building a Smart Store System

Musk is a store keeper and he was tasked to manually record down the store temperature and humidity every half an hour in order to preserve the quality of the goods storing inside. The temperature of the store must be kept below **30°C** and humidity below **70%**. There are currently no **indicators/alerts** in place to alert Musk. Hence, Musk has to manually and regularly monitor the temperature and humidity levels, and in the event if the **temperature or humidity** becomes too high (i.e. over the given threshold levels), Musk has to call the facilities department to activate/repair the air conditioner. Once the temperature and humidity are back to normal, Musk is required to take **a photo of the store** for record purposes.

The current process seems to be too tedious for Musk. You are to help him by **creating a system using raspberry pi, GrovePi+, any available sensors, actuators and camera to improve the efficiency.**

You are required to submit the python code(s) to Luminus.

Project 2: Building a Smart “Robotic” System

Sensors help robots to collect data from the environment, enabling them to react according to certain changes in the environment. You are to **create a robotic system (using raspberry pi, GrovePi+, selected sensors and actuators)** based on the condition table below:

Dark (bright < 50)	Nearby obstacles (D < 20cm)	Button (True/False)	Green LED Status (Robot “Action” 1)	Blue LED Status (Robot “Action” 2)	Red LED Status (Robot “Action” 3)
Yes <u>AND</u>	Yes <u>AND</u>	True	False	False	Blink
No	Yes	False	False	False	True
Yes <u>OR</u>	No	True	False	True	False
No	No	False	True	False	False

Note that we will not be using real robots; using LED to represent different robot actions

You are required to submit the python code(s) to Luminus.